ChatGPT

Capstone Project Plan: Person-Centric Spitting-Event Detection in Railway Surveillance

Executive summary and research positioning

Finalized paper title: A Person-Centric Spatiotemporal Framework for Spitting-Event Detection in Railway Surveillance

This capstone will build an end-to-end computer-vision system that receives railway-platform or CCTV-style video, detects every visible person, maintains a track identity for each person through time, converts each track into short temporal clips, and determines whether that specific individual performs a spitting action. The intended output is therefore not merely *"spitting exists somewhere in this frame"* but a structured event of the form `Track ID → spitting → start time → end time → bounding box → confidence → evidence clip`. Person detection will use a frozen pretrained Ultralytics YOLOv8 detector, tracking will use ByteTrack, and action recognition will be experimentally compared using a frame-level ResNet50 baseline, a ResNet50+LSTM temporal baseline, and a fine-tuned pretrained VideoMAE model. ByteTrack was explicitly designed to recover trajectories by associating both high- and lower-confidence detections, while VideoMAE was designed as a data-efficient self-supervised video representation learner; these are established components rather than contributions of this project. [1]

The proposed dataset is deliberately small but controlled: **exactly 20 retained synthetic source videos**, balanced between sources containing spitting and sources containing hard-negative actions, plus **preferably five, minimum three, staged real clips recorded on a railway platform with permission and consent**. Synthetic human-action data is an established research strategy, including prior work that studies viewpoint diversity, synthetic-to-real domain shift, and rare human events; notably, recent work has also evaluated synthetic rare-event footage using VideoMAE. [2] The synthetic videos therefore are **not themselves the novelty**. They are a transparent data-scarcity solution.

A literature sanity check is important here because spitting detection itself is **not unprecedented**. A 2022 project called *Swachh AI* described YOLOv5-based real-time spitting detection, a 2026 *Swachh Alert* paper describes image/frame-oriented public-spitting detection and dashboarding, and a 2025 urban-hygiene system reported a YOLOv8 approach for smoking, spitting, and urination. [3] General surveillance literature also already contains multi-person action recognition by extracting individual person sequences and spatiotemporal action localization. [4] Consequently, the paper must **not** claim to invent spitting detection, tracking-assisted action recognition, YOLO, ByteTrack, ResNet, LSTM, or VideoMAE.

The defensible research contribution is the **complete identity-aware formulation**: a railway-oriented surveillance framework that decomposes a crowded video into individual tracks, performs temporal spitting recognition per track, explicitly evaluates confusing mouth/head actions, aggregates repeated window predictions into one event, and reports *who, when, and where* rather than only a frame or scene label. The planned small real-platform set further creates an external synthetic-to-real stress test, although three to

five clips are far too few to justify claims of operational railway accuracy. Railway computer-vision surveillance is itself an active area, including systems for people/object detection and tracking; therefore, the railway context strengthens the application motivation but should not be claimed as a completely new surveillance setting. [5]

The project also fits the previously discussed ICMLDE scope: the conference material explicitly lists **Human Activity Recognition** and **Computer Vision**, while its broader tracks include deep CNNs, recurrent networks and sequence modelling. The conference also directs authors to Elsevier's generative-AI policy.
filecite
turn0file0

Abstract — 177 words

Automated detection of spitting in railway environments is challenging because the action is brief, visually subtle, and poorly represented in public surveillance datasets. This capstone proposes a person-centric spatiotemporal framework that first detects people in video, associates individuals across frames using multi-object tracking, and then performs action recognition on track-specific temporal clips. A controlled dataset of 20 synthetic CCTV-style source videos will cover spitting, multi-person scenes, viewpoint and illumination changes, and confusable negatives including drinking, coughing, sneezing, eating, face touching, mouth wiping, bending, and ordinary platform activity. Three to five permissioned, staged railway-platform clips will be held out as an external domain stress test. ResNet50, ResNet50-LSTM, and pretrained VideoMAE will be compared under an identical source-level evaluation protocol. Window-level predictions will be temporally aggregated into events reporting track identity, time interval, spatial location, and confidence. Evaluation will emphasize precision, recall, F1-score, false positives per minute, hard-negative errors, actor attribution, and inference latency. The contribution is an identity-aware end-to-end event-detection framework and evaluation protocol for a scarce public-hygiene action rather than a new backbone architecture.

Keywords: Spitting Detection; Human Action Recognition; Railway Surveillance; Multi-Object Tracking; VideoMAE; Synthetic Video Data.

Paper positioning paragraph: This paper should be positioned as a **systems and application contribution under rare-event data scarcity**, not as a new neural architecture: it proposes an identity-aware, person-centric spatiotemporal pipeline for railway surveillance that converts multi-person footage into tracked individual video sequences, experimentally compares a spatial representation, a conventional sequential representation, and a pretrained video-transformer representation under the same protocol, performs temporal event localization and actor attribution, and explicitly challenges the detector with visually confusable hard negatives and a small staged real-platform domain test. Prior literature already establishes YOLO-based spitting systems, person-level action recognition, ByteTrack, VideoMAE, synthetic human-action data, and railway surveillance, so the defensible novelty lies in the **particular problem formulation, track-wise temporal recognition pipeline, hard-negative protocol, event aggregation, and railway-oriented evaluation**, not in any single component or in an unsupported “first-ever” claim. [6]

Dataset blueprint and generation protocol

The central dataset decision should be **20 retained synthetic source videos, not 20 generated attempts**. For each specification below, it is acceptable to generate up to three candidates and retain the first one that passes quality control; rejected candidates must be logged but must not silently enter training. All retained

videos should ideally use the **same generator, same model/version, same broad visual style, and the same technical generation settings**, while changing seeds, subjects, actions, camera viewpoints, crowd levels, lighting and occlusion. This makes the source of variation auditable. Elsevier's current journal policy expressly permits AI tools used as part of research methodology when their use is described reproducibly, including the tool/model, version and developer or manufacturer; AI-assisted coding should likewise be disclosed in the Methods section. [7]

The reason for devoting half of the source set to negatives is methodological rather than cosmetic. A spitting classifier that sees only spitting versus ordinary standing can learn shortcuts such as forward head motion, a hand close to the mouth, or bending. The dataset must therefore force it to discriminate spitting from actions with similar spatial or temporal cues. Existing synthetic-action literature also demonstrates why viewpoint, body appearance, background and domain shift deserve deliberate control rather than random generation. [8]

Common technical specification for every synthetic source

Property	Final specification
Number retained	Exactly 20
Target duration	9–12 seconds each
Shot	One continuous shot; no edits
Camera	Fixed surveillance/CCTV viewpoint
Aspect ratio	16:9
Requested resolution	1080p where generator supports it; 720p acceptable
Requested frame rate	24–30 fps; normalize all sources to 25 fps in preprocessing
Camera motion	None
Primary labels	`spitting`, `non_spitting`
Negative subtype metadata	`drink`, `cough`, `sneeze`, `eat`, `wipe_mouth`, `face_touch`, `bend_pickup`, `mask_adjust`, `talk_laugh`, `normal_activity`
Setting	Generic South Asian railway-platform-like environment; no actual station name needed
Main action	Once per video, at a defined interval
Synthetic identity	Fictional; no celebrity or recognizable real-person likeness
Visible text/brands	Avoid or make unreadable
Saliva	Do not require visible fluid; classify the bodily action pattern
Audio	Not used

Property	Final specification
Random seed	Record where generator exposes one
Accepted clip QC	Camera stable; identity stable; requested action clear; no severe morphing; no accidental positive action in negative clips

Exact synthetic source specification

ID	Class	Exact scene and action	Main subject / clothing	Duration & action time	People	CCTV geometry	Lighting / crowd / occlusion
S01	Positive	Person waits stationary, glances down, makes one brief head/ neck-forward spitting gesture toward floor, returns upright.	Adult, medium build; blue shirt, dark trousers	10 s; spit **4.2–5.0 s**	1	3.2 m high, frontal-oblique, ~35° downward, subject ~6 m away	Daylight; empty background; none
S02	Positive	Person walks slowly along platform, stops briefly, turns head sideways and performs one spitting gesture, then continues.	Adult, slim build; green kurta, black trousers	10 s; **5.0–5.8 s**	2	3.5 m high, side-oblique, ~40° down	Daylight; one passer-by; low occlusion
S03	Positive	Seated person rises slightly from bench, leans forward and spits toward ground, then settles.	Adult, stocky build; beige shirt, grey trousers	10 s; **4.5–5.4 s**	3	2.8 m high, diagonal ~45°	Daylight; two background people; mild intermittent occlusion

ID	Class	Exact scene and action	Main subject / clothing	Duration & action time	People	CCTV geometry	Lighting / crowd / occlusion
S04	Positive	Standing person slightly bends torso and performs short downward spitting gesture without using hands.	Adult, average build; dark jacket, light trousers	9 s; **3.8–4.7 s**	2	4.0 m high, diagonal, ~45° down	Fluorescent evening/low light; sparse crowd; none
S05	Positive	Five-person platform scene; designated actor standing among moving passengers performs one subtle spit while others continue normally.	Adult, medium build; maroon jacket, dark jeans	12 s; **5.6–6.5 s**	5	4.5 m high wide-angle, diagonal	Daylight; moderate crowd; moderate pedestrian occlusion
S06	Positive	Main actor is relatively small/far in frame, waits and performs one clear head-forward/ downward spitting gesture.	Adult, tall/slim; white shirt, navy trousers	12 s; **6.0–6.9 s**	4	6.0 m high wide shot; actor ~12 m away	Daylight; low crowd; far-field scale

ID	Class	Exact scene and action	Main subject / clothing	Duration & action time	People	CCTV geometry	Lighting / crowd / occlusion
S07	Positive	Actor is briefly crossed by another person, becomes visible again and then performs the spit; tracking continuity is important.	Adult, average build; mustard shirt, dark trousers	11 s; **6.0–6.8 s**	4	3.5 m high, side-oblique	Daylight; moderate crowd; moderate occlusion before event
S08	Positive	Side-profile actor stands near bench, turns slightly away from camera and performs one lateral/ downward spit gesture.	Adult, slim; grey hoodie/ jacket, black trousers	9 s; **4.0–4.8 s**	2	3.0 m high, mostly lateral ~70°	Evening platform lighting; sparse; none
S09	Positive	Actor walks diagonally toward camera, pauses, rotates upper body sideways, spits toward ground, then walks away.	Adult, medium build; pale yellow shirt, dark jeans	10 s; **5.2–6.1 s**	3	3.5 m high, front-oblique	Daylight; mild crowd; brief low occlusion

ID	Class	Exact scene and action	Main subject / clothing	Duration & action time	People	CCTV geometry	Lighting / crowd / occlusion
S10	Positive + hard-negative co-occurrence	Main actor spits while a second visible actor drinks from a bottle approximately one second later; other people wait/ walk.	Spitting actor: dark green shirt; drinking actor: light blue shirt	12 s; spit **5.0–5.9 s**; drink **6.0–7.5 s**	5	4.0 m high wide diagonal	Daylight; moderate crowd; mild-to-moderate occlusion
S11	Negative	Main actor raises bottle to mouth, tilts head, drinks, lowers bottle and continues waiting. **No spitting.**	Adult; red T-shirt, dark trousers	10 s; drink **4.0–6.0 s**	2	3.2 m high side-oblique	Daylight; sparse; none
S12	Negative	Actor coughs twice into bent elbow/ hand, with forward head movement. **No spitting.**	Adult; navy shirt, beige trousers	10 s; cough **4.0–5.5 s**	2	3.5 m high frontal-oblique	Daylight; sparse; none
S13	Negative	Actor prepares to sneeze, covers nose/ mouth, bends head sharply forward once, recovers. **No spitting.**	Adult; light green shirt, dark trousers	9 s; sneeze **3.8–4.8 s**	3	3.3 m high diagonal	Daylight; low crowd; mild partial occlusion

ID	Class	Exact scene and action	Main subject / clothing	Duration & action time	People	CCTV geometry	Lighting / crowd / occlusion
S14	Negative	Actor eats a small snack, raises hand repeatedly near mouth and chews while stationary. **No spitting.**	Adult; brown jacket, grey trousers	10 s; eat **3.5–6.5 s**	2	3.0 m high side view	Daylight; sparse; none
S15	Negative	Actor wipes lips/mouth with tissue or handkerchief, briefly leans head down, then pockets tissue. **No spitting.**	Adult; cream shirt, dark jeans	10 s; wipe **4.2–5.8 s**	3	4.0 m high diagonal	Daylight; low crowd; mild occlusion
S16	Negative	Actor scratches chin, touches lower face and briefly places hand near mouth while looking downward. **No spitting.**	Adult; blue jacket, tan trousers	10 s; face touch **3.8–6.0 s**	2	3.2 m high front-oblique	Evening light; sparse; none
S17	Negative	Actor bends forward/down to pick up a small dropped object and straightens. **No hand-to-mouth and no spitting.**	Adult; white T-shirt, dark trousers	10 s; bend/ pick **4.0–5.5 s**	3	4.5 m high wide diagonal	Daylight; low crowd; slight occlusion

ID	Class	Exact scene and action	Main subject / clothing	Duration & action time	People	CCTV geometry	Lighting / crowd / occlusion
S18	Negative	Actor repeatedly brings hand near lower face to adjust a mask/scarf and then lowers hand. **No spitting.**	Adult; charcoal jacket, light trousers, scarf/ mask	10 s; adjust **4.0–6.0 s**	3	3.5 m high side-oblique	Fluorescent low light; low crowd; none
S19	Negative	Actor talks animatedly to companion, laughs, and briefly moves head/upper torso forward several times. **No spitting.**	Adult; orange shirt, dark jeans	10 s; talk/ laugh **3.0–7.0 s**	4	4.2 m high wide shot	Daylight; moderate crowd; intermittent mild occlusion
S20	Negative	General platform scene: six people walk, wait, check phone and converse; no one drinks, coughs, sneezes or spits.	Mixed fictional adults, neutral varied clothing	12 s; normal throughout	6	4.8 m high wide diagonal	Daylight; moderate crowd; moderate crossing occlusion

The deliberate mixture in **S10** is particularly useful because it tests the central person-centric hypothesis: the pipeline should associate the spitting score with the correct ByteTrack ID rather than simply label the whole scene positive.

Real railway-platform holdout

Five clips are strongly preferable. These are **external stress-test clips, not training data**. They should be filmed only after obtaining appropriate institutional/station/railway permission and written consent from every staged participant. India Cine Hub identifies the Ministry of Railways among central-government stakeholders involved in shooting permissions and states that location permits are handled for particular jurisdictions, dates and locations; domestic applicants can also approach the relevant authorities directly. [9]

ID	Class	Staged clip	Duration	People	Camera	Purpose
R01	Positive	Consenting actor stands normally, then performs a convincing downward **spitting gesture without expelling saliva**, recovers.	8–10 s	1	Fixed, elevated/ oblique where permissible	Basic synthetic→real positive
R02	Positive	Consenting actor walks 2–3 m, pauses/ turns slightly and performs the same gesture, then leaves.	10–12 s	1–2	Fixed, wider platform view	Moving actor + tracking
R03	Negative hard	Actor drinks from water bottle, with head tilt and hand-to-mouth motion.	8–10 s	1–2	Same general view	Drinking false-positive test
R04	Negative hard	Actor coughs into elbow and subsequently wipes mouth.	8–10 s	1–2	Same general view	Cough/wipe false-positive test
R05	Negative normal	Three consenting participants wait, walk, talk/check phones; no spitting.	10–15 s	3	Fixed wide view	Multi-person false alarms and tracking

If permissions or practical constraints limit collection to **three** real clips, retain **R01, R03 and R05**. Do not replace them with random public surveillance recordings.

Three to five clips do **not** constitute a statistically meaningful real-world test set. In the paper, these results should be reported individually as an **external staged domain stress test** rather than as "real-world accuracy."

Master synthetic-video generation prompt

Every S01–S20 prompt should be derived mechanically from the following template and the parameter table above. Store the exact filled prompt in `source_metadata.csv`.

```
Generate one continuous fixed-surveillance video for a human-action-recognition
research dataset.

TECHNICAL REQUIREMENTS
- Duration: {DURATION_SECONDS} seconds.
```

```
- Aspect ratio: 16:9.
- Resolution: 1920x1080 if supported, otherwise highest available >=720p.
- Frame rate: 24–30 fps if selectable.
- One uninterrupted shot.
- Fixed surveillance/CCTV camera only.
- Absolutely no camera pan, tilt, zoom, dolly, handheld shake, cuts,
transitions,
  slow motion, time lapse, rack focus, or cinematic camera movement.

ENVIRONMENT
Create a generic South Asian railway-platform-like environment with plausible
platform flooring, benches/columns and railway-station architecture. Do not show
a readable real station name, company logo, advertisement, brand, license plate,
or identifiable real person. The environment should look like ordinary
surveillance footage rather than a movie scene.

CAMERA
- Camera height: {CAMERA_HEIGHT_M} metres.
- View direction: {CAMERA_AZIMUTH}.
- Downward pitch: approximately {CAMERA_PITCH_DEG} degrees.
- Main-subject distance: approximately {SUBJECT_DISTANCE_M} metres.
- Keep camera geometry constant for the full clip.

MAIN SUBJECT
A fictional adult {SUBJECT_BUILD} person wearing {CLOTHING_DESCRIPTION}.
Preserve the same face, body proportions, clothes, accessories and identity in
every frame. The subject must remain visually consistent.

BACKGROUND
There are {TOTAL_PEOPLE} visible people in total. Background people perform only
ordinary railway-platform behaviours such as standing, waiting, walking,
looking at phones or talking unless another action is explicitly specified.
Crowd density: {CROWD_DENSITY}.
Occlusion condition: {OCCLUSION_DESCRIPTION}.
Lighting condition: {LIGHTING_DESCRIPTION}.

TIMELINE
From 0.0 seconds until {ACTION_START_S} seconds, the main subject behaves
normally.
From {ACTION_START_S} to {ACTION_END_S} seconds, the subject performs exactly
this action:

{ACTION_DIRECTIVE}

After {ACTION_END_S} seconds, the main subject returns to neutral ordinary
behaviour until the video ends.

MOTION QUALITY
```

```
The action should be anatomically realistic, subtle and clearly visible from the
specified surveillance distance. Preserve temporal continuity of head, hands,
torso and legs. People must not teleport, duplicate, disappear, exchange
clothing or change identity.

RESEARCH CONSTRAINTS
- Do not exaggerate the action for cinematic effect.
- Do not create a close-up of the mouth.
- Do not show graphic bodily fluid.
- Do not add captions, labels, arrows, bounding boxes or subtitles.
- Do not imitate any known public figure.
- Keep rails, flooring, benches, columns and architecture geometrically stable.

NEGATIVE CONSTRAINTS
No cuts. No moving camera. No face morphing. No extra limbs. No fused hands.
No disappearing people. No duplicated pedestrians. No warped railway geometry.
No floating objects. No sudden clothing changes. No text overlays. No cinematic
depth-of-field. No dramatic lighting changes. No unintended spitting action by
background people.
```

Exact action directives to insert

IDs	`ACTION_DIRECTIVE`
S01	`The person briefly lowers the chin, presses the lips, makes a short forward head-and-neck motion consistent with spitting toward the ground, and immediately recovers. Do not show coughing, drinking, vomiting or exaggerated fluid.`
S02	`While walking, the person pauses, turns the head approximately 45 degrees sideways, makes one short head-and-neck-forward spitting gesture toward the ground, then resumes walking.`
S03	`From the seated position, the person leans the upper torso forward, performs one brief downward spitting gesture using head and mouth motion, then returns toward the seated posture.`
S04	`The person makes a shallow torso bend and one brief downward head-and-neck spitting gesture, then straightens. Hands remain away from the mouth.`
S05	`The designated main subject, while surrounded by ordinary pedestrians, performs one subtle but visible downward spitting gesture and returns immediately to neutral; every other person continues ordinary movement and does not spit.`
S06	`The distant main subject makes one clearly readable but natural head-forward/downward spitting gesture. The action must remain visible despite the subject being relatively small in the frame.`

IDs	ACTION_DIRECTIVE
S07	After a pedestrian finishes partially occluding the main subject, the main subject performs one short downward spitting gesture and returns to neutral. Preserve the same identity before and after occlusion.
S08	In side profile, the person turns slightly away, performs one brief lateral-and-downward spitting gesture using head and neck motion, and returns upright.
S09	The person pauses walking, rotates the upper body slightly sideways, performs one downward/sideways spitting gesture, then resumes walking.
S10	The main subject performs one downward spitting gesture at the specified interval. Approximately one second later, a different clearly distinguishable person raises a bottle and drinks from it. Keep both people visible and preserve their separate identities.
S11	The person raises a bottle to the lips, tilts the head slightly, swallows for roughly two seconds and lowers the bottle. The person must NOT spit before, during or after drinking.
S12	The person coughs twice into a bent elbow or hand, with realistic brief forward head movements. There must be NO spitting and NO fluid expulsion.
S13	The person brings a hand toward the nose/mouth, makes one realistic covered sneeze with a rapid forward head movement, then recovers. Do NOT show spitting.
S14	The person eats a small snack, brings one hand near the mouth several times and chews naturally. The person must NOT spit or cough.
S15	The person uses a tissue or handkerchief to wipe the lips and lower face, briefly lowers the head and then puts the tissue away. Do NOT show spitting.
S16	The person scratches the chin, touches the lower face and briefly keeps a hand close to the mouth while looking down. Do NOT show spitting, drinking or coughing.
S17	The person bends forward to pick up a small object from the floor, reaches downward, retrieves it and straightens. Hands do not approach the mouth and no spitting occurs.
S18	The person raises both or one hand near the lower face to adjust a mask or scarf, holds it briefly, then lowers the hand. Do NOT show spitting.
S19	The person talks animatedly to a companion, laughs and makes several natural forward head/torso motions during conversation. No person spits.

IDs	`ACTION_DIRECTIVE`
S20	`All people perform ordinary waiting, walking, phone checking and conversation only. No one spits, drinks, coughs, sneezes, eats, wipes the mouth or performs an exaggerated mouth-related motion.`

Generation variables

For each row S01–S20, take `duration`, subject/clothing, people count, camera height/direction, lighting, crowd/occlusion and action timestamps directly from the source specification table. Do not let the generator creatively change those variables. Generate candidate `C1`; inspect it manually. Generate `C2` or `C3` only when necessary.

Synthetic quality-control acceptance checklist

A candidate is accepted only when the camera remains fixed, the requested action occurs once at approximately the requested time, the main person's identity/clothing remains stable, hands/head are not severely malformed during the action, no pedestrian teleports or duplicates conspicuously near the event, no negative video accidentally contains spitting-like expulsion, the action remains visible at surveillance scale, and the scene has no severe railway-geometry corruption. A failed candidate receives a rejection reason such as `camera_motion`, `action_missing`, `identity_morph`, `wrong_action`, `accidental_positive`, `severe_occlusion`, `geometry_error`, or `other`.

This design follows the broader research lesson that synthetic action data can be useful but carries a synthetic-to-real domain gap; the external staged clips exist specifically to reveal, not hide, that gap. [10]

Framework architecture and experimental design

The framework is deliberately split into **localization/tracking** and **action understanding**. This is essential to the paper's person-centric formulation. YOLO answers *where are the people?*, ByteTrack answers *which detection belongs to the same person over time?*, and the action model answers *what is this tracked person doing during this temporal window?* ByteTrack's original method associates detections over time and explicitly recovers useful lower-score detections that would otherwise fragment trajectories. [11] Ultralytics' tracking interface supports ByteTrack through a dedicated tracker configuration, making the combination practical in one Python inference stack. [12]

Complete inference flow

```
flowchart TD
    A[Railway / CCTV Video] --> B[Decode and normalize to 25 FPS]
    B --> C[YOLOv8s Person Detection<br/>class = person only]
    C --> D[ByteTrack Multi-Object Tracking]
    D --> E1[Track ID 1]
    D --> E2[Track ID 2]
    D --> E3[Track ID N]
```

```
E1 --> F[Per-track padded ROI buffer]
E2 --> F
E3 --> F

F --> G[2.0 s windows<br/>0.5 s stride]
G --> H[Uniformly sample 16 frames<br/>224 × 224]

H --> I1[ResNet50<br/>Spatial baseline]
H --> I2[ResNet50 + LSTM<br/>Temporal baseline]
H --> I3[VideoMAE<br/>Primary video model]

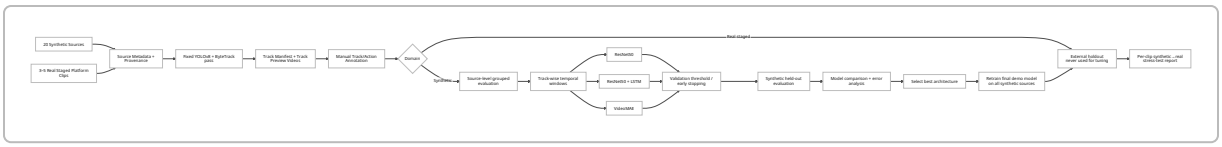
I1 --> J[Window probability p(spitting)]
I2 --> J
I3 --> J

J --> K[3-window temporal smoothing]
K --> L[Validation-selected threshold]
L --> M[Require consecutive positive windows]
M --> N[Merge events on same track<br/>gap <= 1.0 s]

N --> O[Spitting Event]
O --> P[Track ID]
O --> Q[Start / end time]
O --> R[Bounding box / location]
O --> S[Confidence]
O --> T[Evidence clip]

P --> U[CSV + JSON + Annotated MP4]
Q --> U
R --> U
S --> U
T --> U
U --> V[Streamlit Review Dashboard]
```

Dataset-to-model flow



Detection and tracking

Use `yolov8s.pt` as the fixed person detector. The detector is not fine-tuned for the capstone initially; it is merely the spatial front end. Use COCO `person` class only. YOLOv8 is intentionally frozen as a project dependency even though the Ultralytics ecosystem has evolved further; changing detectors midway would

change track identities and invalidate manually created track annotations. Ultralytics supports detection, pose and tracking in the same codebase and provides ByteTrack as a supported tracker. [13]

Initial detector configuration:

```
detector:
  weights: yolov8s.pt
  classes: [0]            # COCO person
  imgsz: 640
  conf: 0.25
  iou: 0.50
  device: cuda
```

Tracking configuration:

```
tracker:
  implementation: ultralytics_bytetrack
  config: bytetrack.yaml
  persist_tracks: true
  minimum_track_duration_s: 2.0
  roi_padding_fraction: 0.18
```

The 18% padding avoids cutting off head/arms because spitting cues can include upper-torso posture and hand-mouth interaction. Bounding boxes are clipped to image limits after expansion.

A critical data-engineering rule is that **ByteTrack's numerical ID is only an internal trajectory identifier, not a person's civil identity**. The project must never attempt face recognition, name identification or re-identification across unrelated videos.

Before labeling actions, run detector + tracker once with the finalized configuration and save a `tracking_run_id`. Generate a short preview clip for every resulting track. Human annotators then label those tracks and their action intervals. If tracking parameters are materially changed later, invalidate and regenerate these annotations because `track_id=4` is not guaranteed to represent the same trajectory after a tracking rerun.

Temporal clip extraction

Every input video is normalized to **25 fps**. For every sufficiently long person track:

1. Maintain a continuous **2.0-second context window**, equivalent to 50 normalized source frames.
2. Advance the window by **0.5 seconds**, approximately 12–13 frames.
3. Uniformly sample **16 frames** from the 50-frame interval.
4. Resize/preprocess the padded person ROI to **224 × 224**.
5. Feed exactly the same 2-second temporal span to every model.

The 16-frame, 224-pixel representation is especially natural for VideoMAE because its standard configuration uses 16 frames and 224 × 224 images. [14]

For the ResNet50 baseline, only the **center frame** of the 2-second window is provided to the network. This is intentional: it tests whether temporal evidence adds value.

For ResNet50+LSTM and VideoMAE, all 16 sampled frames are used.

Positive and negative windows must not be labeled solely from source-level labels. Use event annotations:

```
source_id
track_id
action_label
action_start_s
action_end_s
actor_role
```

A window is `spitting` when the manually annotated spitting interval substantially overlaps the window. Windows immediately at ambiguous action boundaries may be marked `ignore` instead of forcing a potentially incorrect binary label.

Action-recognition models

ResNet50 spatial baseline. ResNet established residual learning as a standard deep visual representation and provides a strong frame-level control. [15] Use ImageNet-pretrained torchvision weights, replace the classification head with two output classes, and train on the center frame.

```
Center frame
     ↓
ResNet50
     ↓ 2048-D
Dropout 0.3
     ↓
Linear(2048 → 2)
     ↓
Spitting / Non-spitting
```

ResNet50 + LSTM temporal baseline. Each of the 16 frames passes through a shared pretrained ResNet50 feature extractor. The resulting temporal sequence of 2048-dimensional vectors is sent to a two-layer LSTM. This provides a conventional “spatial CNN + sequence model” baseline:

```
f01 —→ ResNet —→ x01 ┐
f02 —→ ResNet —→ x02 │
```

```
...                    ├→ 2-layer LSTM → Dropout → Linear → p(spitting)
f16 —→ ResNet —→ x16 ┘
```

Use a **unidirectional** LSTM rather than bidirectional LSTM so the architecture remains closer to a deployable causal interpretation, although predictions are still emitted after a complete two-second window.

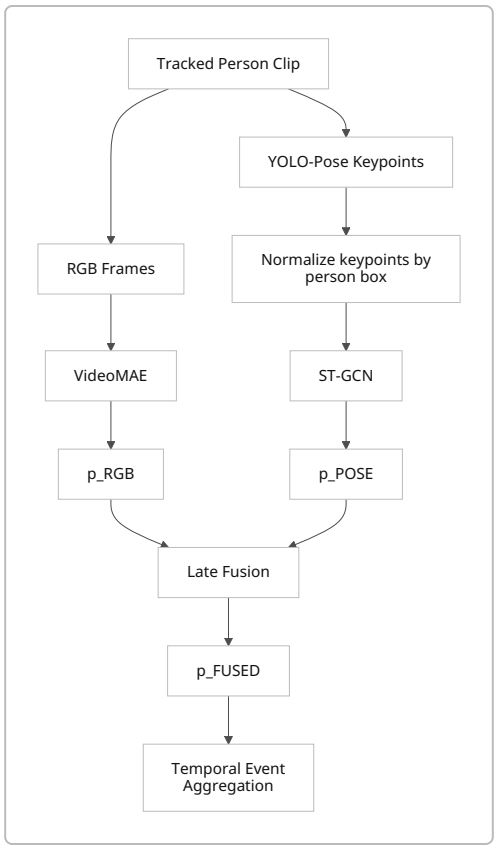
VideoMAE. VideoMAE will be the strongest planned RGB-video model, but it must be described as a **fine-tuned pretrained architecture**, not "our proposed neural network." The original work introduced masked-autoencoder pretraining for video and reported good data efficiency; the official Transformers implementation exposes a 16-frame, 224 × 224 configuration and supports video classification. [16]

```
16 × 224 × 224 person frames
             ↓
Pretrained VideoMAE
             ↓
Classification head
             ↓
p(spitting)
```

Start from `MCG-NJU/videomae-base` or its appropriate video-classification checkpoint and **fine-tune rather than training from scratch**. The project's source dataset is far too limited to support meaningful from-scratch transformer training; transfer learning is fundamental to the experiment.

Optional pose branch

Treat pose as a **stretch objective, not a prerequisite for completing the capstone**. The preferred implementation is **YOLOv8-Pose**, because the project already uses Ultralytics. OpenPose is an alternative; however, adding a second large pose ecosystem creates more integration overhead. ST-GCN is a well-established architecture for learning spatial and temporal patterns over skeleton joints. [17]



Use a simple, interpretable late fusion:

$$p_{\text{fused}} = \alpha p_{\text{RGB}} + (1 - \alpha) p_{\text{pose}}$$

Test `α ∈ {0.5, 0.7, 0.8}` **only on validation data**. Do not add pose if the RGB core has not already been completed and evaluated.

Event aggregation

Raw window scores should not directly produce alerts. A single physical spit may trigger several overlapping windows.

For each `track_id`:

```
2.0–4.0 s   p=.09
2.5–4.5 s   p=.18
3.0–5.0 s   p=.61
3.5–5.5 s   p=.87
4.0–6.0 s   p=.94
4.5–6.5 s   p=.89
```

```
5.0–7.0 s   p=.52
5.5–7.5 s   p=.17
```

Processing:

```
raw probabilities
        ↓
median filter over 3 consecutive windows
        ↓
validation-selected threshold τ
        ↓
require ≥2 consecutive positive windows
        ↓
merge same-track candidates separated by ≤1.0 s
        ↓
one event
```

Event confidence should be the **mean of the three highest smoothed scores** inside the merged event, or the mean of all if fewer than three exist.

Example final record:

```
{
  "source_id": "S05",
  "track_id": 3,
  "event_type": "spitting",
  "start_s": 4.96,
  "end_s": 6.48,
  "confidence": 0.91,
  "peak_bbox_xyxy": [612, 284, 792, 694],
  "model": "videomae",
  "evidence_clip": "outputs/evidence/S05_track003_event001.mp4"
}
```

Evaluation protocol and ablations

Because 20 sources are very few, a single random clip-level train/test split would be especially misleading. Multiple windows extracted from the same source are strongly correlated. The finalized protocol should therefore be **source-grouped five-fold evaluation**.

Define each synthetic source as either `contains_spit=true` or `false`: S01–S10 versus S11–S20. Use **five outer folds**, each holding out **four complete source videos: two positive-containing sources and two negative sources**. Every track, frame and window from a source travels with that source. Within the remaining 16 sources of each fold, designate four entire sources, again two positive and two negative, as

validation; the remaining 12 are training. This produces 12 train / 4 validation / 4 outer-test sources per fold, and every synthetic source serves as a held-out outer test source exactly once. Scikit-learn provides grouped/stratified utilities suitable for implementing such source-aware evaluation; the planned environment pins scikit-learn 1.9.0. [18]

Never do this:

```
S05 window #1 → train
S05 window #2 → validation
S05 window #3 → test
```

Correct:

```
S05 and every derived track/window → exactly one partition in a fold
```

The 3–5 real clips remain **outside all synthetic folds** and must never be used for threshold selection, model choice or early stopping.

Report mean ± standard deviation over the five synthetic outer folds for:

Metric	Why it matters
Precision	Fraction of alerts that are actually spitting
Recall	Fraction of annotated spitting events found
F1-score	Main balanced classification summary
Balanced accuracy	Less sensitive to class imbalance
PR-AUC	Useful when positive windows are relatively rare
False positives per minute	Operationally interpretable surveillance metric
Event precision/recall/F1	Evaluates merged physical events, not duplicated windows
Track-attribution accuracy	Whether correct person ID receives the event
Hard-negative confusion	Drinking/coughing/etc. failure analysis
Inference FPS	Computational throughput
End-to-end latency	Delay from input window to event result

For event matching, predeclare a **temporal IoU of at least 0.5** between predicted and ground-truth action intervals as a correct localization. Because the test is small, also show timing error in seconds rather than hiding it behind one threshold.

The experiments should answer specific research questions:

Experiment	Research question
ResNet50 vs ResNet50+LSTM	Does temporal context outperform a single image for spitting?
ResNet50+LSTM vs VideoMAE	Does modern pretrained video representation improve temporal recognition?
Full-frame VideoMAE vs track-crop VideoMAE	Does person-centric decomposition improve recognition in multi-person scenes?
Raw threshold vs temporal aggregation	Does event merging reduce duplicate/unstable alerts?
Hard-negative subtype analysis	What actions most commonly mimic spitting?
Synthetic outer-fold vs staged real clips	How severe is the initial synthetic→real domain gap?
Optional VideoMAE vs VideoMAE+pose	Does skeleton motion provide complementary evidence?

The **whole-frame versus track-wise VideoMAE ablation** is particularly important because it directly tests the main framework claim. If track-wise recognition does not outperform or otherwise improve offender attribution compared with whole-scene classification, the paper must report that honestly.

After all cross-validation experiments are finished and the winning architecture is selected, train one **final demonstration checkpoint** using all 20 synthetic sources with the hyperparameters fixed from the synthetic experiments. Use the median selected epoch/threshold configuration rather than tuning anything against R01–R05. Then freeze that checkpoint and run the external platform clips once.

Technology stack and training specification

The software stack should be intentionally conservative and reproducible rather than automatically using every newest release. PyTorch's official installation archive documents PyTorch 2.12.1 paired with torchvision 0.27.1 and provides CUDA 12.6/13.x wheels. [19] Hugging Face Transformers 5.13.1 was published in July 2026 and supports Python 3.10+ and PyTorch 2.4+; its VideoMAE documentation provides the model interface needed here. [20] OpenCV Python 4.12.0.88 is a mature prebuilt release suitable for video I/O rather than requiring the newly released OpenCV 5 line. [21] Streamlit 1.60.0 was released on July 21, 2026. [22] Pandas 3.0.4 should specifically **not** be selected because PyPI records it as yanked for reported datetime-related segfaults; use 3.0.5 instead. [23]

Pinned core environment

Component	Pin / choice	Role
OS	Ubuntu 24.04 LTS preferred; Windows 11 + WSL2 acceptable	Development environment

Component	Pin / choice	Role
Python	**3.11.15**	Runtime
PyTorch	**2.12.1**	Deep-learning framework
torchvision	**0.27.1**	ResNet50 + image transforms
CUDA wheel	**12.6** initially	GPU PyTorch runtime
Ultralytics	**8.4.52**	YOLOv8 detection / tracking / pose integration
YOLO weights	`yolov8s.pt`	Person detection
ByteTrack	Ultralytics `bytetrack.yaml`	Multi-object tracking
Hugging Face Transformers	**5.13.1**	VideoMAE
VideoMAE checkpoint	`MCG-NJU/videomae-base`	Pretrained video backbone
OpenCV-Python	**4.12.0.88**	Decode, resize, crop, annotate, export
scikit-learn	**1.9.0**	Metrics and source-aware split utilities
pandas	**3.0.5**	Metadata/event tables
Streamlit	**1.60.0**	Demo dashboard
Matplotlib	Resolve compatible version and lock	Evaluation plots
PyYAML	Resolve and lock	Config files
Pillow	Resolve and lock	Images/transforms
tqdm	Resolve and lock	CLI progress
pytest	Resolve and lock	Unit/integration tests

The selected Ultralytics release is a **pinned reproducibility choice**, not a claim that it is the newest release; Ultralytics' repository supports detection, tracking and pose, and the published tracking interface supports ByteTrack configuration. [24] The open-source Ultralytics repository is AGPL-3.0; that is normally compatible with an academic/student open research workflow, but licensing must be re-evaluated before turning the capstone into a proprietary commercial product. [25]

After the environment is tested, Codex should generate both a human-readable requirements file and a full `requirements-lock.txt` / `pip freeze` snapshot. The lockfile, Python version, CUDA version, GPU name and git commit hash must accompany the final experiments.

Recommended hardware

A practical target is an NVIDIA GPU with **12 GB or more VRAM**, 32 GB system RAM, and at least 50 GB free SSD storage. An RTX 3060 12 GB, RTX 4070 12 GB or 4060 Ti 16 GB-class machine is sufficient as a capstone development target when VideoMAE uses small physical batches and gradient accumulation. If only 8 GB VRAM is available, reduce physical VideoMAE batch size to 1–2 and use gradient accumulation. CPU is entirely adequate for metadata manipulation and some video preprocessing but should not be the planned VideoMAE training environment.

Preprocessing

```
video:
  normalized_fps: 25
  window_seconds: 2.0
  stride_seconds: 0.5
  sampled_frames: 16
  image_size: 224

track_filter:
  min_duration_seconds: 2.0
  bbox_padding: 0.18
  min_bbox_height_px_at_720p: 80  # heuristic; log rejected tracks
```

The minimum bounding-box size is a **starting heuristic**, not a scientifically established constant. Record how many tracks it removes and revisit it only using training/validation data.

Common augmentations

Use conservative training-only spatial augmentation:

```
augmentation:
  random_resized_crop: true
  crop_scale: [0.85, 1.0]
  horizontal_flip_probability: 0.5
  mild_brightness_contrast: true
  max_rotation_deg: 5
```

Apply one spatial transformation consistently across all frames of a temporal clip; do **not** independently flip/crop each frame, which would manufacture false temporal motion. Do not use aggressive temporal reversal because spitting has directional temporal structure.

ResNet50 baseline hyperparameters

```
model: resnet50
pretrained: imagenet
input: center_frame
```

```
image_size: 224
num_classes: 2

head:
  dropout: 0.30

stage_1:
  freeze_backbone: true
  epochs: 3
  optimizer: AdamW
  lr_head: 0.001
  weight_decay: 0.0001

stage_2:
  unfreeze: [layer4, fc]
  max_epochs: 22
  lr_backbone: 0.00001
  lr_head: 0.0001
  weight_decay: 0.0001

batch_size: 16
loss: weighted_cross_entropy_if_needed
early_stopping:
  metric: validation_f1
  patience: 5
seed: 42
amp: true
```

ResNet50 + LSTM hyperparameters

```
model: resnet50_lstm
frames: 16
image_size: 224
feature_dim: 2048

lstm:
  hidden_size: 256
  num_layers: 2
  bidirectional: false
  dropout: 0.30

classifier_dropout: 0.30
num_classes: 2

optimizer: AdamW
lr_resnet_layer4: 0.00001
```

```
lr_lstm_and_head: 0.0001
weight_decay: 0.0001
batch_size: 8
max_epochs: 25
early_stopping_patience: 5
seed: 42
amp: true
```

VideoMAE hyperparameters

The default VideoMAE configuration documented by Hugging Face uses 16 frames, 224 × 224 images, patch size 16 and tubelet size 2, which matches the proposed window representation. [14]

```
model: videomae
checkpoint: MCG-NJU/videomae-base
frames: 16
image_size: 224
num_classes: 2

fine_tuning:
  initial_trainable:
    - classifier
    - final_4_transformer_blocks
  optional_full_unfreeze_after_epoch: 4

optimizer: AdamW
lr_backbone: 0.00002
lr_head: 0.0001
weight_decay: 0.05

physical_batch_size: 2
gradient_accumulation_steps: 4
effective_batch_size: 8

max_epochs: 20
warmup_fraction: 0.10
early_stopping:
  metric: validation_f1
  patience: 4

gradient_clip_norm: 1.0
seed: 42
amp: true
```

If fine-tuning all four final transformer blocks proves unstable, train only the classifier plus final two blocks first. Do not compensate for instability by repeatedly selecting hyperparameters on the outer test fold.

Training sampling

Because an 11-second source with a 0.5-second stride can generate many near-duplicate windows, the training loader should not allow one long track to dominate simply because it yields more windows. Store all windows, but use a source/class-balanced sampler or impose a reproducible per-track sampling cap per epoch. This addresses correlation; it does not magically convert hundreds of derived windows into hundreds of independent source videos.

Event configuration

```
event_aggregation:
  smoothing: median
  smoothing_window_count: 3
  threshold:
    method: maximize_validation_f1
  min_consecutive_positive_windows: 2
  merge_gap_seconds: 1.0
  event_confidence: mean_top_3
```

The threshold must be estimated separately inside each fold from validation data. Never choose it after inspecting outer test or staged-real performance.

Expected exported artifacts

```
outputs/
├── checkpoints/
│   ├── resnet50/
│   ├── resnet_lstm/
│   └── videomae/
├── metrics/
│   ├── fold_01/
│   ├── fold_02/
│   ├── ...
│   ├── cross_validation_summary.csv
│   └── staged_real_stress_test.csv
├── plots/
│   ├── confusion_matrix_*.png
│   ├── precision_recall_*.png
│   ├── hard_negative_breakdown.png
│   └── model_comparison.png
├── annotated/
│   └── *.mp4
├── evidence/
│   └── *_track*_event*.mp4
```

```
├── events/
│   ├── *.csv
│   └── *.json
└── logs/
```

The dashboard should show the uploaded/input video, selected checkpoint, processing progress, annotated playback, a table of merged spitting events, confidence, track ID and timestamp, and buttons for downloading CSV, JSON and the annotated video. For the scientific experiment, the dashboard threshold must default to the frozen validation-derived threshold, even if a manual slider exists for demonstration.

Codex implementation specification

Coding paragraph: Using Codex, the project should be implemented as a modular Python/PyTorch application in which OpenCV performs video decoding, frame-rate normalization, cropping and annotation; a pinned Ultralytics YOLOv8s model detects the COCO person class and ByteTrack maintains per-video track IDs; a preprocessing subsystem expands each person's bounding box, creates per-track buffers, generates two-second windows every 0.5 seconds and uniformly samples 16 frames; torchvision ResNet50 implements the spatial baseline and supplies per-frame embeddings to a two-layer LSTM for the sequential baseline, while Hugging Face Transformers fine-tunes VideoMAE for track-level video classification; an optional YOLO-Pose/ST-GCN branch can later provide skeleton probabilities for late fusion; shared training and evaluation code enforces complete source-level isolation, records reproducible fold manifests and computes window-, event-, hard-negative- and actor-attribution metrics; inference smooths model scores, thresholds them, merges adjacent same-track detections into one event and exports annotated MP4, evidence clips, CSV and JSON; a lightweight Streamlit application provides review and downloads; all configuration is YAML-driven, metadata/provenance is stored in CSV/JSON, random seeds and package versions are locked, and automated tests verify split isolation, frame/window sampling, cropping, tracking interfaces, event merging and end-to-end inference.

The following block is the **implementation prompt to give Codex together with this report**.

```
You are implementing a CAPSTONE research project. Treat the attached report as
the source of truth. Do not silently change the scientific protocol. If a
requirement is infeasible because of an API/library difference, document the
reason and make the smallest compatible change.

PROJECT NAME
A Person-Centric Spatiotemporal Framework for Spitting-Event Detection in
Railway Surveillance

PRIMARY OBJECTIVE
Create a reproducible Python/PyTorch research pipeline that:
1. accepts railway/CCTV-style MP4 source videos;
2. detects all persons with pretrained Ultralytics YOLOv8s;
3. tracks persons with ByteTrack;
4. creates person-specific temporal tracks and windows;
5. trains/evaluates:
```

```
   a. ResNet50 frame baseline,
   b. ResNet50 + LSTM temporal baseline,
   c. pretrained VideoMAE fine-tune;
6. optionally supports YOLO-Pose + ST-GCN as a stretch branch;
7. converts window probabilities into merged spitting events;
8. outputs Track ID + event start/end + bbox + confidence;
9. exports annotated MP4, evidence clips, events CSV and events JSON;
10. provides a simple Streamlit review dashboard.

SCIENTIFIC CONSTRAINTS
- Never fabricate dataset files, labels, metrics, checkpoints or results.
- Never write placeholder accuracy values as if experiments were completed.
- Missing data must produce an informative error.
- A source video and every track/window derived from it must NEVER exist in more
  than one train/validation/test partition during a fold.
- Real staged platform clips must NEVER be used for model selection, threshold
  tuning or early stopping.
- Track IDs are transient trajectory IDs only; implement NO face recognition.
- ResNet50, ResNet50+LSTM and VideoMAE must use the same source folds.
- The ResNet50 baseline sees the center frame of the same 2.0-second window.
- LSTM and VideoMAE see the same 16 uniformly sampled frames.
- VideoMAE is pretrained/fine-tuned, never trained from scratch.
- All thresholds are selected using validation data only.
- Save all configs and resolved environment versions with every run.
- Set and record seed=42 for Python, NumPy and PyTorch.
- Use mixed precision when CUDA is available.
- CPU fallback must work for smoke tests even if it is slow.

TARGET ENVIRONMENT
Python 3.11.15
torch==2.12.1
torchvision==0.27.1
ultralytics==8.4.52
transformers==5.13.1
opencv-python==4.12.0.88
scikit-learn==1.9.0
pandas==3.0.5
streamlit==1.60.0

Use compatible pinned versions of numpy, scipy, matplotlib, pyyaml, pillow,
tqdm and pytest. After the environment is working, create
requirements-lock.txt from the actual resolved environment.

Prefer Ubuntu / WSL2 and CUDA 12.6 PyTorch wheels.
Code must also have a CPU mode.

CREATE THIS PROJECT STRUCTURE
```

```
project/
├── README.md
├── pyproject.toml
├── requirements.txt
├── requirements-lock.txt
├── .gitignore
├── configs/
│   ├── data.yaml
│   ├── detector.yaml
│   ├── tracker.yaml
│   ├── resnet50.yaml
│   ├── resnet_lstm.yaml
│   ├── videomae.yaml
│   ├── pose_stgcn.yaml
│   └── inference.yaml
├── data/
│   ├── raw/
│   │   ├── synthetic/
│   │   └── real_staged/
│   ├── metadata/
│   │   └── source_metadata.csv
│   ├── annotations/
│   │   └── action_intervals.csv
│   ├── tracking/
│   │   ├── manifests/
│   │   └── previews/
│   ├── processed/
│   │   ├── tracks/
│   │   └── windows/
│   └── splits/
├── src/
│   ├── __init__.py
│   ├── common/
│   │   ├── config.py
│   │   ├── logging_utils.py
│   │   ├── reproducibility.py
│   │   ├── schemas.py
│   │   └── video_io.py
│   ├── data/
│   │   ├── validate_sources.py
│   │   ├── normalize_video.py
│   │   ├── extract_tracks.py
│   │   ├── make_track_previews.py
│   │   ├── validate_annotations.py
│   │   ├── build_folds.py
│   │   ├── build_windows.py
```

```
│   │   ├── datasets.py
│   │   ├── samplers.py
│   │   └── transforms.py
│   ├── models/
│   │   ├── registry.py
│   │   ├── resnet50_classifier.py
│   │   ├── resnet_lstm.py
│   │   ├── videomae_classifier.py
│   │   └── pose_stgcn.py
│   ├── train/
│   │   ├── engine.py
│   │   ├── train_resnet50.py
│   │   ├── train_resnet_lstm.py
│   │   └── train_videomae.py
│   ├── evaluation/
│   │   ├── metrics.py
│   │   ├── event_metrics.py
│   │   ├── evaluate_fold.py
│   │   ├── cross_validation.py
│   │   └── error_analysis.py
│   ├── inference/
│   │   ├── track_buffer.py
│   │   ├── classifier_runner.py
│   │   ├── event_aggregator.py
│   │   ├── renderer.py
│   │   └── pipeline.py
│   └── dashboard/
│       └── app.py
├── scripts/
│   ├── 01_validate_sources.py
│   ├── 02_normalize_sources.py
│   ├── 03_extract_tracks.py
│   ├── 04_make_track_previews.py
│   ├── 05_build_folds.py
│   ├── 06_build_windows.py
│   ├── 07_train_cross_validation.py
│   ├── 08_evaluate_cross_validation.py
│   ├── 09_train_final_model.py
│   ├── 10_test_real_staged.py
│   └── 11_run_inference.py
├── tests/
│   ├── test_metadata_schema.py
│   ├── test_source_split_no_leakage.py
│   ├── test_window_sampling.py
│   ├── test_bbox_padding.py
│   ├── test_track_buffer.py
```

```
│   ├── test_event_aggregator.py
│   ├── test_metrics.py
│   └── test_end_to_end_smoke.py
└── outputs/
    ├── checkpoints/
    ├── metrics/
    ├── plots/
    ├── annotated/
    ├── evidence/
    ├── events/
    └── logs/

DATASET INPUT CONTRACT

Synthetic source files:
data/raw/synthetic/S01.mp4 ... S20.mp4

Real staged files:
data/raw/real_staged/R01.mp4 ... R05.mp4
Allow R04/R05 to be absent when only 3 clips are collected.

SOURCE METADATA CSV REQUIRED COLUMNS
source_id
file_name
domain
contains_spit
primary_label
negative_subtype
generator_provider
generator_model
generator_version
generation_date
prompt_id
prompt_text
seed
candidate_index
qc_status
qc_rejection_reason
location_type
permission_reference
consent_reference
fps_original
fps_normalized
width
height
duration_s
main_actor_count
```

```
crowd_count
camera_height_m
camera_angle
lighting
occlusion_level
split_role
sha256

Use blank/not_applicable generator fields for real staged videos.

ACTION ANNOTATION CSV REQUIRED COLUMNS
source_id
tracking_run_id
track_id
actor_role
action_label
action_start_s
action_end_s
negative_subtype
annotation_confidence
notes

VALIDATE:
- source ID is unique;
- referenced file exists;
- duration and FPS are sane;
- action interval is inside source duration;
- source/track pair exists in tracking manifest;
- primary labels belong to known enumeration;
- real clips cannot be assigned to synthetic training folds.

VIDEO NORMALIZATION
- Normalize all raw videos to 25 FPS.
- Preserve original raw video.
- Do not stretch playback duration.
- Store normalized file under processed or a dedicated normalized directory.
- Log original/normalized dimensions and FPS.
- Calculate SHA-256 for raw source.

PERSON DETECTION
Use:
from ultralytics import YOLO
weights = "yolov8s.pt"

Person class only.
Initial:
imgsz=640
conf=0.25
```

```
iou=0.50

Make detector configurable.
Do NOT train YOLO in the core implementation.

TRACKING
Use Ultralytics ByteTrack with bytetrack.yaml.
Persist identity across frames.
Store for every frame:
source_id
frame_index
timestamp_s
track_id
x1
y1
x2
y2
det_conf

Save one Parquet or CSV manifest per source.
Generate per-track preview MP4s showing the box and Track ID.
Store a tracking_run_id derived from detector/tracker config hash.
If config changes, tracking_run_id changes.

TRACK FILTERS
- min track duration 2.0 s
- starting minimum bbox height at 720p-equivalent = 80 px
- bbox padding = 18 percent of width/height
- clamp padded bbox to frame boundary
- never write invalid/zero-size crops
- log rejected tracks and reasons

WINDOWING
Normalize source to 25 FPS.
Window length = 2.0 s = 50 normalized source frames.
Stride = 0.5 s.
From every 2 s window uniformly sample exactly 16 frame indices.
Return:
pixel data / file paths
source_id
track_id
start_s
end_s
16 selected frame indices
label
negative subtype
fold ID
```

```
If a tracked person is temporarily absent:
- interpolate only tiny bounding-box gaps if tracker output permits safely;
- otherwise mark incomplete window and reject it;
- do NOT hallucinate frames.

Create a deterministic uniform sampling function and unit test it.

RESNET50 DATASET
Input = center sampled/canonical frame from each 2 s window.
224 x 224.
Use torchvision pretrained ResNet50.
Final head:
Dropout(0.3)
Linear(2048, 2)

RESNET50 TRAINING
Stage 1:
backbone frozen
3 epochs
AdamW
head lr=1e-3
weight_decay=1e-4

Stage 2:
unfreeze layer4 + fc
max 22 epochs
backbone lr=1e-5
head lr=1e-4
weight_decay=1e-4
batch 16
early stop on validation F1, patience 5

RESNET-LSTM MODEL
16 frames per sample.
Shared ResNet50 encodes each frame to 2048-D feature.
2-layer unidirectional LSTM:
input_size=2048
hidden_size=256
num_layers=2
dropout=0.3

Final:
last valid hidden representation
Dropout(0.3)
Linear(256, 2)

Train:
batch=8
```

```
AdamW
ResNet layer4 lr=1e-5
LSTM/head lr=1e-4
weight_decay=1e-4
max epochs=25
patience=5

VIDEOMAE MODEL
Use Hugging Face Transformers.
Checkpoint:
MCG-NJU/videomae-base

Video classification with 2 labels:
non_spitting = 0
spitting = 1

Input:
16 RGB frames
224 x 224

Initial fine tune:
classification head + final 4 transformer blocks
Optional full-unfreeze after epoch 4 if validation is stable.

Optimizer:
AdamW
backbone lr=2e-5
head lr=1e-4
weight_decay=0.05
physical batch size=2
gradient accumulation=4
effective batch=8
max epochs=20
warmup=10 percent
gradient clip=1.0
early stop validation F1 patience=4
AMP on CUDA

Do not train from scratch.

TRANSFORMS
Training only:
- RandomResizedCrop scale 0.85–1.0
- horizontal flip p=0.5
- mild brightness/contrast
- rotation <=5 degrees

CRITICAL:
```

```
for video models, apply the SAME random spatial transform parameters to all
16 frames in a clip. Do not independently augment each frame.

No temporal reversal.

DATA LEAKAGE / FOLDS
There are 20 synthetic source IDs.
S01-S10 contain a spitting event.
S11-S20 contain no spitting event.

Build five deterministic outer folds.
Each outer test fold contains:
2 contains-spit source videos
2 no-spit source videos

Each source is outer-test exactly once.

Within each outer fold, from the remaining 16 sources:
- validation = 4 complete sources (2 positive-containing, 2 negative)
- train = remaining 12 sources.

ALL windows and tracks from a source remain together.

Write fold manifests:
data/splits/fold_01_train.txt
data/splits/fold_01_val.txt
data/splits/fold_01_test.txt
...
fold_05_*.txt

A test must assert intersections are empty.

Real R01-R05:
external_real.txt
never appear in any synthetic fold.

TRAINING SAMPLER
Do not let a long source dominate because it produces many overlapping windows.
Implement source/class-balanced sampling.
Report number of source videos, tracks and windows separately.
Never describe windows as independent source videos.

METRICS
Window level:
- precision
- recall
- F1
- balanced accuracy
```

```
- PR-AUC
- confusion matrix

Event level:
- precision
- recall
- F1
- temporal IoU >=0.5 match
- start/end timing error

Surveillance:
- false positives per minute on no-spit source videos
- track attribution accuracy for multi-person positives
- FPS
- end-to-end latency

Hard-negative:
output confusion / false-positive counts by:
drink
cough
sneeze
eat
wipe_mouth
face_touch
bend_pickup
mask_adjust
talk_laugh
normal_activity

Cross-validation:
save per-fold CSV/JSON and aggregate mean + standard deviation.
Never invent missing metrics.

EVENT AGGREGATION
Input:
timestamped per-window p(spitting) per track.

Algorithm:
1. sort by track_id and window time;
2. median-smooth probability across 3 consecutive windows;
3. threshold tau is chosen from VALIDATION data to maximize F1;
4. require at least 2 consecutive above-threshold windows;
5. merge candidate events on same track when temporal gap <=1.0 s;
6. event start/end = union of candidate windows;
7. confidence = mean of top 3 smoothed probabilities inside event;
8. event peak bbox = bbox at window with maximum smoothed score.

Expose this as pure functions/classes so it can be unit-tested.
```

```
EVENT CSV
source_id
video_file
event_id
track_id
event_type
start_s
end_s
duration_s
confidence
peak_x1
peak_y1
peak_x2
peak_y2
model_name
checkpoint
evidence_clip_path

Also output equivalent JSON.

INFERENCE PIPELINE
CLI example:

python scripts/11_run_inference.py \
  --input path/to/video.mp4 \
  --model videomae \
  --checkpoint outputs/checkpoints/final_videomae.pt \
  --output-dir outputs/demo_run

Pipeline:
decode
detect people
track
maintain per-track rolling buffer
construct 2-s windows / 0.5-s stride
sample 16 frames
classify
aggregate events
render boxes:
- neutral tracks: ID only
- detected event: ID + "SPITTING" + confidence
write annotated video
write evidence clip for each event
write CSV + JSON

Do not hard-code color names/styles into scientific logic.
```

```
FULL-FRAME ABLATION
Implement an evaluation mode where VideoMAE sees the same source frames without
person cropping. This is needed for:
whole-frame VideoMAE vs track-wise VideoMAE.

Do NOT let this ablation block initial core development.

OPTIONAL POSE BRANCH
Feature flag:
enable_pose: false

When enabled:
- use YOLOv8 pose model
- extract 2D keypoints per tracked person
- map pose sequence to same track/time window
- normalize x/y by bounding box dimensions and confidence-mask missing joints
- implement ST-GCN as an optional module
- output p_pose
- late fuse:
  p_fused = alpha*p_rgb + (1-alpha)*p_pose
- alpha choices 0.5, 0.7, 0.8 selected on validation only.

The optional pose branch must NEVER prevent the RGB system from working.

FINAL MODEL
After cross-validation:
- choose architecture based on predeclared metrics, primarily mean event F1 and
  false positives/minute;
- fix hyperparameters;
- derive a final epoch count/threshold policy from synthetic CV only;
- train final demo model using all 20 synthetic sources;
- freeze checkpoint;
- then evaluate R01-R05 ONCE.

REAL STAGED REPORTING
Do not report "real-world accuracy" from 3–5 clips.
Output one row per real source:
predicted event?
correct track?
start/end?
confidence?
false alerts?
notes?

DASHBOARD
Streamlit app:
- page title and project summary
- upload MP4 OR choose a local demo file
```

```
- select model/checkpoint
- display fixed scientific threshold from checkpoint metadata
- optional demo threshold override clearly labelled "demo only"
- process button
- progress/status
- annotated video playback
- event table
- evidence clip links/previews
- download annotated MP4
- download CSV
- download JSON

Do not display names/face identities.
Only transient Track IDs.

REPRODUCIBILITY
Every training run directory must save:
config snapshot
git commit if repository exists
Python version
torch/torchvision version
CUDA version
GPU name
all package versions
seed
fold manifest
training log
best checkpoint
best epoch
validation threshold
metrics

Store checkpoint metadata next to weight file.

LOGGING
Use Python logging.
No print-only pipelines.
Log INFO for normal progress.
Log WARNING for recoverable corrupt tracks/frames.
Raise descriptive exceptions for invalid metadata/splits.

TESTS

test_metadata_schema.py
- valid metadata accepted
- missing required columns rejected
- invalid labels rejected
```

```
test_source_split_no_leakage.py
- no source appears in train+val/test together
- every S01-S20 is outer test exactly once
- real clips never enter folds

test_window_sampling.py
- 2-s/25fps expected raw window length
- returns exactly 16 monotonically ordered frame indices
- deterministic

test_bbox_padding.py
- 18% expansion
- boxes remain within frame
- zero-size boxes rejected

test_track_buffer.py
- windows emitted on correct stride
- IDs never mix

test_event_aggregator.py
- isolated one-window spike does not create event
- two consecutive positives can create event
- same-track candidates <=1s merge
- >1s gap creates two events
- different track IDs never merge
- confidence calculation correct

test_metrics.py
- compare simple hand-computable prediction arrays
- false positives per minute correct

test_end_to_end_smoke.py
- programmatically create tiny dummy video
- mock detector/tracker/classifier where necessary
- run pipeline
- verify CSV/JSON and output video are created

README
Write a comprehensive README containing:
- scientific overview
- pipeline diagram using Mermaid
- environment setup
- PyTorch CUDA 12.6 installation note
- dataset placement
- metadata schema
- annotation workflow
- preprocessing commands
- training commands
```

```
- CV evaluation commands
- inference command
- dashboard command
- test command
- reproducibility notes
- ethics/privacy note
- troubleshooting

CONFIG DRIVEN
No scientifically important magic numbers buried in code.
Window duration, stride, threshold, merge gap, model hyperparameters and paths
must come from YAML/config objects.

QUALITY
Use type hints.
Use pathlib rather than string path concatenation.
Use dataclasses/Pydantic-style validation where appropriate without creating
unnecessary complexity.
Docstrings for public functions/classes.
Avoid giant scripts; separate IO, modeling, training, metrics and visualization.
Make CLI scripts thin wrappers over src modules.

DEVELOPMENT ORDER
Do NOT attempt everything at once.

MILESTONE A
Scaffolding + config + metadata validator + tests.

MILESTONE B
Video normalization + YOLO person detection + ByteTrack + track previews.

MILESTONE C
Action annotation validator + source folds + window generation + leakage tests.

MILESTONE D
ResNet50 baseline training/evaluation.

MILESTONE E
ResNet50+LSTM training/evaluation.

MILESTONE F
VideoMAE fine-tuning/evaluation.

MILESTONE G
Window-score inference + event aggregation + output renderer.

MILESTONE H
Five-fold cross-validation runner + comparison tables + error analysis.
```

```
MILESTONE I
Final model training + staged-real stress-test pipeline.

MILESTONE J
Streamlit dashboard.

MILESTONE K
Optional pose/ST-GCN only if A-J are stable.

ACCEPTANCE CRITERIA
The core project is complete when:
1. pytest passes;
2. raw video can be converted to per-track temporal windows;
3. all three required model families can train on identical fold manifests;
4. no source leakage is possible through normal APIs;
5. five-fold evaluation generates real metrics from real model predictions;
6. full-video inference produces stable Track IDs and timestamped scores;
7. event aggregation produces one event rather than many overlapping alerts;
8. inference exports MP4 + CSV + JSON + evidence clips;
9. Streamlit can display those outputs;
10. the external real staged clips can be processed without entering training;
11. all experiments save configs/environment/seeds;
12. no code claims a metric that was not actually computed.

When implementing, prioritize a correct, reproducible core over clever
optimizations. The optional pose branch is strictly secondary.
```

Ethics, permissions, provenance and reproducibility

The 3–5 platform recordings must be treated as **staged research recordings**, not casually collected public surveillance. India Cine Hub's current framework states that filming permissions can be location-specific and identifies the Ministry of Railways among central-government stakeholders; for a student capstone, the operational route may differ from commercial filming, so the team should obtain a written determination or permission from the relevant station/railway/institution rather than assuming that small-scale filming is automatically exempt. [26]

The Digital Personal Data Protection Act, 2023 concerns processing of digital personal data while recognizing both individual protection and lawful processing purposes. A recognizable video of a bystander can therefore create privacy considerations; the project should minimize collection rather than rely on a broad assertion that public-space video is harmless. [27] This report is not a legal determination, so the conservative capstone procedure is: written actor consent, local/institutional authorization, no minors, no covert collection, camera framing that avoids uninvolved passengers, and blurring or discarding footage containing identifiable nonparticipants where practical.

Do **not** require actors to genuinely expel saliva on a railway platform. Stage the head/neck/mouth motion. This is scientifically consistent with the proposed detector because it is intended to learn a human action sequence rather than microscopic saliva droplets, and it avoids deliberately creating the public-hygiene behavior the project is meant to study.

The synthetic-video methodology must be fully transparent. Elsevier's current journal policy states that when AI tools form part of research design or methods they should be described reproducibly, including model/tool name, version and developer/manufacturer where applicable, and that AI-assisted code development should be declared as part of the research methodology. [7] That directly applies to both **synthetic video generation** and **Codex-assisted software development**.

A suitable paper Methods disclosure would be:

> Synthetic CCTV-style video data were generated using [provider/model/version] under controlled prompts specifying action, camera geometry, subject appearance, crowd density, lighting and temporal action location. Up to three candidate generations were permitted per planned source specification, with one retained following predefined manual quality-control criteria. Prompts, generation metadata, rejected-candidate reasons and source hashes were preserved for provenance. AI-assisted software development using Codex was employed during implementation; all generated code was reviewed, tested and experimentally validated by the authors.

Do not cite an AI generator as an “author.” Cite its technical paper/official model documentation as appropriate and describe its methodological use. Elsevier also makes authors accountable for checking tool terms, privacy and intellectual-property implications. [7]

Required source metadata schema

```
source_id
file_name
domain                      # synthetic | real_staged
contains_spit               # true | false
primary_label
negative_subtype

generator_provider
generator_model
generator_version
generation_date
prompt_id
prompt_text
seed
candidate_index
qc_status
qc_rejection_reason
```

```
location_type
permission_reference
consent_reference

fps_original
fps_normalized
width
height
duration_s

main_actor_count
crowd_count
camera_height_m
camera_angle
lighting
occlusion_level

split_role
sha256
```

Required track/action annotation schema

```
source_id
tracking_run_id
track_id
actor_role
action_label
action_start_s
action_end_s
negative_subtype
annotation_confidence
notes
```

Ethics and provenance gate before experiments

Requirement	Must be true before final paper
Synthetic clips explicitly identified as synthetic	Yes
Generator provider/model/version recorded	Yes
Full prompts retained	Yes
Rejected-generation reasons logged	Yes
Source file hashes recorded	Yes

Requirement	Must be true before final paper
Generator output/data-use terms checked	Yes
Codex use disclosed in Methods if required by publisher policy	Yes
Railway/platform filming authorization verified	Yes
Staged actors consented	Yes
No minors intentionally recorded	Yes
No actual spitting required	Yes
Bystanders minimized	Yes
Identifiable unwanted bystanders blurred/discarded where appropriate	Yes
Real clips excluded from model tuning	Yes
Source-level leakage test passes	Yes
Raw staged clips stored access-controlled	Yes
Publication/release rights separately checked before sharing real footage	Yes

The project should release code and synthetic provenance if the generator's terms permit. Raw real-platform footage does **not** need to be published merely for the sake of openness; publication/release should depend on participant consent, station permission and institutional guidance.

Timeline and capstone deliverables

A realistic plan is **eight core weeks plus two contingency/polish weeks**. The project should be managed around scientific milestones rather than “build dashboard first.” The dashboard is useful for the capstone demonstration but cannot compensate for a leaky or poorly designed experiment.

Week	Core work	Required deliverable
Week A	Freeze research specification; select generator; create metadata templates; obtain/clarify railway filming permission; generate first candidate synthetic clips	Project repository, dataset schema, final prompts, permission request/ documentation
Week B	Complete and QC 20 synthetic sources; normalize videos; implement YOLOv8 + ByteTrack extraction; generate track previews	20 accepted sources, QC log, track manifests and previews

Week	Core work	Required deliverable
Week C	Label track/action intervals; implement source folds; window extraction; build leakage/window unit tests	`action_intervals.csv`, five fold manifests, processed window index
Week D	Implement/train ResNet50; verify metrics and experiment logging; begin hard-negative error analysis	First reproducible baseline results
Week E	Implement/train ResNet50+LSTM under identical folds	Temporal baseline results and comparison
Week F	Integrate/fine-tune VideoMAE; run grouped synthetic evaluation; perform whole-frame versus person-track ablation	Main model results and person-centric ablation
Week G	Implement temporal smoothing/event merging; track-attribution metric; annotated inference; record preferably five staged real-platform clips	End-to-end inference and external holdout dataset
Week H	Freeze winning model; run real staged stress test once; complete error analysis and Streamlit dashboard	Final quantitative tables, real-domain table, demo
Week I	Optional pose/ST-GCN only if core is stable; otherwise improve data/QC, repeat failed runs and optimize documentation	Optional fusion results or stronger core
Week J	Paper/report writing, figures, reproducibility audit, presentation/video demo, final package	Capstone report, paper draft, code, presentation and demo

The **minimum viable capstone** is complete at the end of Week H without pose. Pose is not required for a good project.

The final capstone package should contain: the 20-source synthetic dataset and full provenance; the permissioned staged real test clips or appropriately controlled derivatives; fixed action annotations; source-aware folds; trained ResNet50, ResNet50+LSTM and VideoMAE experiments; cross-validation metrics; the whole-frame/person-track ablation; hard-negative analysis; event aggregation; full-video demo inference; Streamlit dashboard; source code and automated tests; environment lockfile; and the research manuscript.

A strong final results section should be organized around evidence rather than a single accuracy number:

```
Research Question
       │
       ├── Does temporal modelling matter?
       │       ResNet50 vs ResNet50+LSTM
       │
       ├── Does stronger video pretraining matter?
```

```
│       ResNet50+LSTM vs VideoMAE
│
├── Does person-centric decomposition matter?
│       Whole-frame VideoMAE vs Track-wise VideoMAE
│
├── Can the system distinguish confusable actions?
│       Drinking / coughing / sneezing / wiping / etc.
│
├── Can it identify the responsible person?
│       Track attribution in S05/S07/S10 and other multi-person scenes
│
├── Can repeated window detections become one useful alert?
│       Raw scores vs temporal event aggregation
│
└── Does a model learned mainly from synthetic footage survive
        a small real-platform domain shift?
        R01–R05 external stress test
```

The capstone should be considered successful even if VideoMAE does not dominate every baseline or the staged-real performance reveals a sizeable domain gap. A scientifically valuable outcome can be that temporal models reduce certain hard-negative errors, that person tracking enables correct offender attribution even when overall classification remains challenging, or that synthetic-only training fails to transfer adequately and demonstrates exactly where additional real training data is required. Synthetic-to-real domain shift is already recognized as an important issue in action recognition, so quantifying it honestly is more defensible than overstating deployment readiness. [28]

The **final scientific claim**, assuming the experiments support it, should therefore be narrow and defensible:

> **The capstone demonstrates an end-to-end, person-centric spatiotemporal pipeline for detecting and temporally localizing spitting actions at the tracked-individual level in railway-style surveillance video, and evaluates the feasibility and limitations of bootstrapping that pipeline from a controlled synthetic dataset using hard-negative action cases and a small, independently held-out staged railway-platform stress test.**

That is a stronger paper than claiming to have invented a new model, and it accurately distinguishes the project from prior image/YOLO-based public-spitting systems and from generic multi-person action-recognition pipelines. [29]

---

[1] [11] ECVA | European Computer Vision Association
https://www.ecva.net/papers/eccv_2022/papers_ECCV/html/315_ECCV_2022_paper.php?utm_source=chatgpt.com

[2] [8] Synthetic Humans for Action Recognition from Unseen Viewpoints
https://arxiv.org/abs/1912.04070?utm_source=chatgpt.com

3 6 29 Swachh AI: Real-time Spitting Detection using Camera | Peer-reviewed Journal
https://ijarcce.com/papers/swachh-ai-real-time-spitting-detection-using-camera/?utm_source=chatgpt.com

4 Spatio-Temporal Action Detection with Multi-Object Interaction
https://arxiv.org/abs/2004.00180?utm_source=chatgpt.com

5 Automatic Surveillance of People and Objects on Railway Tracks. | International Journal of Interactive Multimedia and Artificial Intelligence
https://www.ijimai.org/index.php/ijimai/article/view/826?utm_source=chatgpt.com

7 Generative AI policies for journals
https://www.elsevier.com/en-au/about/policies-and-standards/generative-ai-policies-for-journals?utm_source=chatgpt.com

9 26 International Projects | India Cine Hub
https://indiacinehub.gov.in/index.php/permission/step-by-step-guide/international-projects?utm_source=chatgpt.com

10 28 Synthetic-to-Real Domain Adaptation for Action Recognition: A Dataset and Baseline Performances
https://arxiv.org/abs/2303.10280?utm_source=chatgpt.com

12 13 Suivi multi-objets YOLO dans les vidéos | Ultralytics
https://docs.ultralytics.com/fr/modes/track?utm_source=chatgpt.com

14 VideoMAE · Hugging Face
https://huggingface.co/docs/transformers/model_doc/videomae?utm_source=chatgpt.com

15 Deep Residual Learning for Image Recognition
https://doi.org/10.1109/CVPR.2016.90?utm_source=chatgpt.com

16 VideoMAE: Masked Autoencoders are Data-Efficient Learners for Self-Supervised Video Pre-Training
https://papers.nips.cc/paper_files/paper/2022/hash/416f9cb3276121c42eebb86352a4354a-Abstract-Conference.html?utm_source=chatgpt.com

17 Spatial Temporal Graph Convolutional Networks for Skeleton-Based Action Recognition - AAAI
https://aaai.org/papers/12328-spatial-temporal-graph-convolutional-networks-for-skeleton-based-action-recognition/?utm_source=chatgpt.com

18 scikit-learn · PyPI
https://pypi.org/project/scikit-learn/

19 Previous PyTorch Versions
https://pytorch.org/get-started/previous-versions/?utm_source=chatgpt.com

20 transformers · PyPI
https://pypi.org/project/transformers/5.13.1/?utm_source=chatgpt.com

21 opencv-python · PyPI
https://pypi.org/project/opencv-python/4.12.0.88/?utm_source=chatgpt.com

22 streamlit · PyPI
https://pypi.org/project/streamlit/?utm_source=chatgpt.com

23 pandas · PyPI
https://pypi.org/project/pandas/?utm_source=chatgpt.com

24 25 GitHub - ultralytics/ultralytics: Ultralytics YOLO 🚀 · GitHub
https://github.com/ultralytics/ultralytics?utm_source=chatgpt.com

27 India Code: Digital Personal Data Protection Act, 2023.
https://www.indiacode.nic.in/handle/123456789/22037?utm_source=chatgpt.com